Week 1: Modeling What Matters

Senegal Case – Retail Access and Fair Optimization

Peter Zhang

12768/19867/94867: From Data to Action

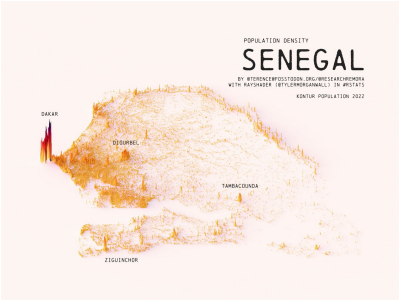
Course Theme: Models and Mirrors

- Every model encodes values.
- The hardest decisions are made before modeling: **What do "you" want?**
- This week's focus:
  - Objective specification
  - Social welfare functions
  - Predict-then-optimize flows
  - Case 1: Senegal retail access
  - Case 2: Hospital surge planning

Where We Are Going



Senegal, westernmost country in Africa.



Population density. The **Senegal River Valley** runs along the northern border – the strip we will spend the hour in.

Dakar



Dakar.



Thiéboudienne – *rice* and fish.

Île de Gorée



Île de Gorée, off Dakar. UNESCO World Heritage site.

- Twenty minutes by ferry from the capital.
- A reminder that in this country, questions about *who gets access to what* are not abstract and not new.

Livelihood: Rice, and the Valley That Grows It

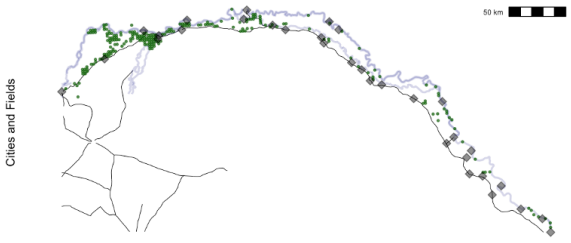


Green: irrigated fields. Grey diamonds: the 32 cities. Blue: rivers. Black: paved roads.

Brennan, Zhang et al.

- 600+ km of valley, six mostly rural states.
- **225** irrigated fields, roughly **1,000** farmers each.
- Average plot: **0.2 hectares**. Half an acre.
- Rice, plus cereals, cash crops and vegetables under forward contract.

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- Rice, plus cereals, cash crops and vegetables under forward contract.
- Agriculture is about 15–17% of GDP and 30% of employment.

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- That window is **one to two days**, and it now opens **5 to 15 times a year**.
- **And the equipment fails.** Pumps last three to five years; pipes fail faster and less predictably.

> When a pump fails inside a two-day window, the farmer's problem is **how far away the nearest shop is**.

The Distance Problem





Two of the surveyed shops; pipes visible on the left.

Brennan, Zhang et al.

- Walking: 2 km/h. Sand road: 17. Pavement: 30. All of it roughly *halves* in the rains — exactly when draining is needed.
- Average farmer: 5 km on foot to any road, then 2.6 km of sand for every 1 km of pavement, then $\sim$3 hours waiting for a bus.

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- Average farmer: 5 km on foot to any road, then 2.6 km of sand for every 1 km of pavement, then $\sim$3 hours waiting for a bus.
- A 34 km round trip for pipes becomes a **fourteen-hour day**.

Elsewhere in African essential services the rules of thumb are two hours to reach health care, three kilometres to reach a school.

How Bad was Status Quo?

	Pumps	Pipes	Parts	Used	Repairs
Mean round trip (km)	36	34	31	32	25
Worst 10% of fields (km)	72	56	54	54	46
Maximum (km)	100	100	114	104	60
Shop in the *nearest* city	52%	40%	57%	43%	71%
Cities (of 32) with such a shop	9	12	13	11	24

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- One farmer in two has no shop selling equipment in their nearest city.

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- Repairs are everywhere. **Goods are not.**
- One farmer in two has no shop selling equipment in their nearest city.
- And farmers in areas the *state* has underserved are also further from *private* retail: 45 km vs. 28 km for pumps. A double burden nobody had measured.

The Measurement Problem

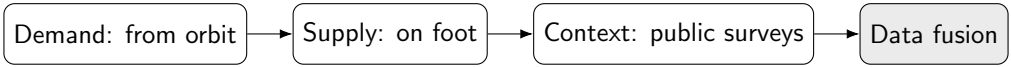
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- There is almost no industry-wide evidence on African retail.
- You cannot optimize a network nobody has ever mapped.

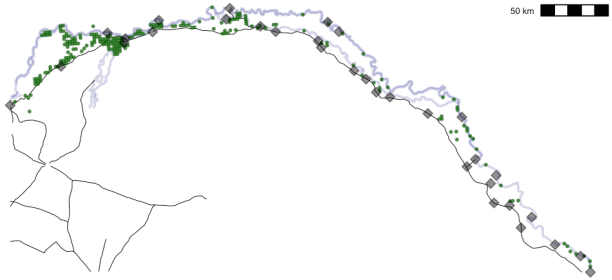
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- There is almost no industry-wide evidence on African retail.
- You cannot optimize a network nobody has ever mapped.
- So the first three years of this project were not modeling. They were **measurement** — and it was only ever *partially* solved.



Measurement I — Demand, From Orbit

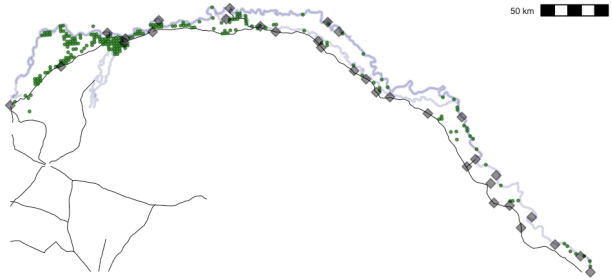


Fields and cities along the Valley.

Brennan, Zhang et al.

- Cannot interview 225,000 farmers to find out where they farm.
- USGS/NASA **Landsat** land-use classification, 2017 imagery.
- **225** parcels in the Valley classified as irrigated.
- Each 200-ha field becomes one demand point.

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- **225** parcels in the Valley classified as irrigated.
- Each 200-ha field becomes one demand point.
- 23 fields were separately ground-surveyed to confirm there is no farm-side retail.

"Predict" in predict-then-optimize did not require a trained model here.

Measurement II — Supply, On Foot



Copyright: Mark Brennan.

- A door-to-door census of *all 32 cities* in the Valley.
- $\sim$**1,100** candidate shops screened by sign, stock and conversation. **145** actually sell or repair irrigation equipment.
- **Ten person-months**, in French, Wolof, Pulaar and Sereer.

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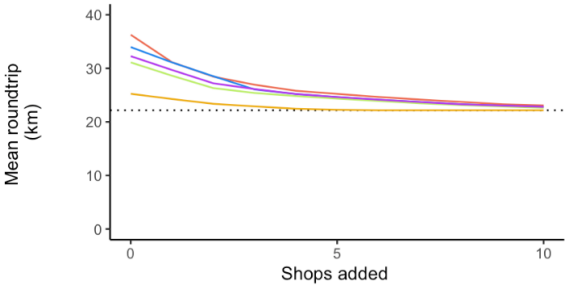
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- $\sim$**1,100** candidate shops screened by sign, stock and conversation. **145** actually sell or repair irrigation equipment.
- **Ten person-months**, in French, Wolof, Pulaar and Sereer.
- The obstacles were not analytical: local licence plates and a donkey cart for social access; camping overnight for physical access; a stamped ministry letter for legal access on a national border.
- Four validity checks, including questions designed to elicit both a *yes* and a *no*.

Measurement III — The Instrument

A slice of the field questionnaire — 173 fields in total, administered in French. Note **Q29**: that one line is the distance parameter in our model.

Q	Question (instrument administered in French)
9	Est-ce que vous pouvez evoir une route à partir de votre plantation ?
10	Es-ce que la route est pavée ou non ?
11	Est-ce que vous pouvez voir le périmètre ou bien l'eau souterraine à partir de votre...
13	Est-ce que la plantation a : (cocher si oui)
14	Depuis quand avez-vous commence à irriguer votre plantation de riz ? (Année)
15	Combien de fois cultivez-vous du riz: une, deux, ou trois fois par an ?
16_1	Combien d'hectares au total est cette parcelle cette saison ? - L'estimation du cultivateur
18_1	Quelle sorte d'irrigation pratiquez-vous ? - Publique
19	Cultivez-vous toujours dans un projet avec [sorte d'irrigation de l'agriculteur] ?
23	Vendez-vous certains produits de vos récoltes ?
26	Etes-vous utilisez une pompe pour irriguer votre riz?
27	Etes-vous propriétaire des POMPES que vous utilisez?
29	De quelle ville avez-vous acheté or reçu votre pompe?
30	Etes-vous propriétaires des TUYAUX utilisées pour l'irrigation du riz?
31	De quelle ville avez-vous acheté ou reçu votre tuyau de canalisation ?
35	Est-ce vous payez actuellement pour l'eau d'irrigation?
37_1	Quelle a été la dernière fois que vous aviez reçu des informations ou des conseils à propos...

So What Can Be Done?

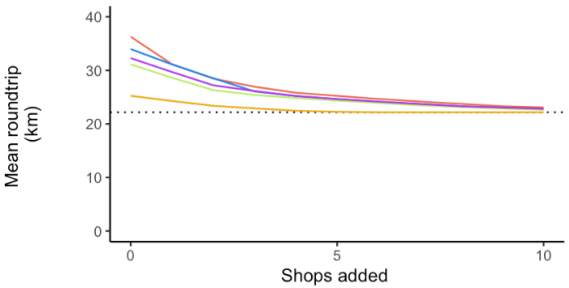


Mean round trip against shops added.

Brennan, Zhang et al., Fig. 4.

- The state does not sell pumps. Since the 1980s, private retail does.
- Levers: tax incentives, land grants, subsidised credit, donor matching grants, information sharing. Any of these can "activate" a city.

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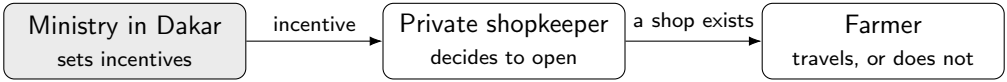
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- Levers: tax incentives, land grants, subsidised credit, donor matching grants, information sharing. Any of these can "activate" a city.
- Budgets are small: recent initiatives funded **5 to 20** facilities.
- Five well-placed shops cut the average trip by about **a third** — more than half of everything that is achievable.

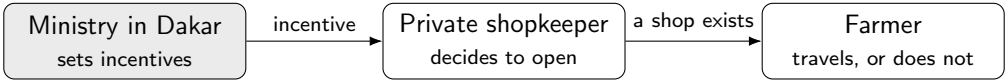
Steep, then flat. Senegal does not need to rebuild its retail sector.

Who Is Making the Decision?



- In the 1970s this was a classical facility location problem: the state owned the facilities and placed them.
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- Today the state owns none of them. It can only make a location attractive.
- **The decision-maker cannot implement its own decision.** Everything downstream is somebody else's choice.

What We *Can* Write Down: the Constraints

$\sum_k y_{ik} = 1$	each field goes to exactly one city
$y_{ik} \leq x_k$	only to a city that has a shop
$x_k \geq b_k$	existing shops stay open (we cannot close them)
$\sum_k x_k \leq \sum_k b_k + c$	budget: at most c new cities
$x_k,\ y_{ik} \in \{0, 1\}$	every decision is yes or no

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- Notice: **every one of these is uncontroversial.** We could get a room of stakeholders to sign all five in ten minutes.

And Now the Hard Part

$$\min\ ?$$

What should this model *minimize*?

Candidate Objectives

- **Average distance.** $\min \sum_i d_i$

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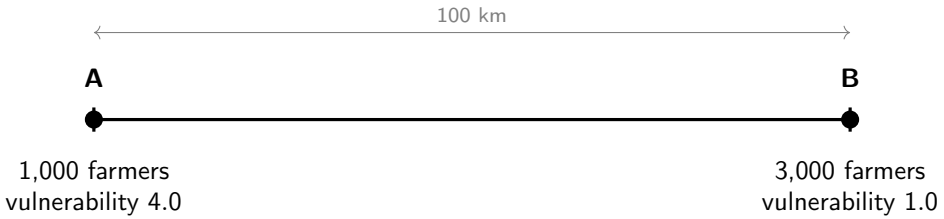
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- **Coverage.** $\min \#\{i : d_i > T\}$

  The form a policy target is actually written in.

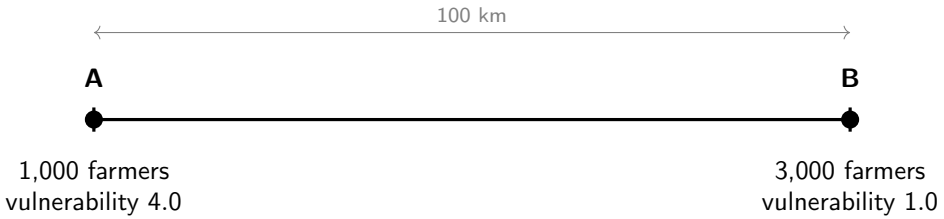
Same data. Same constraints. Same solver. **Different maps.**

Toy Example



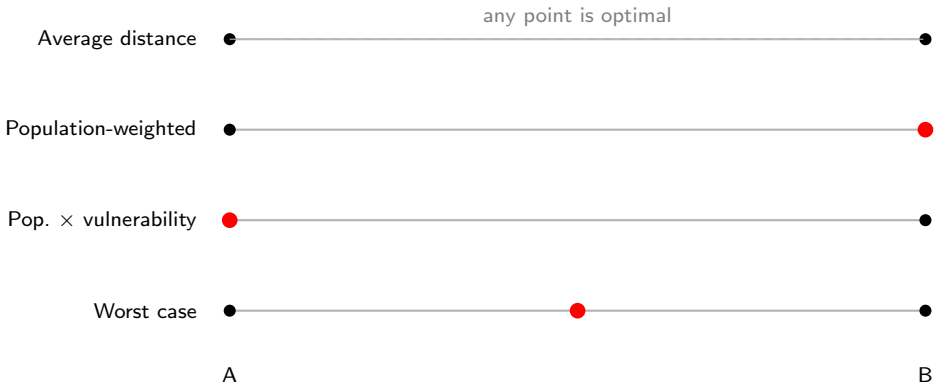
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- Place the shop at position $x \in [0, 100]$.
- Village A is small but badly exposed. Village B is large but better off.
- Weights: $p_A v_A = 1000 \times 4 = \mathbf{4000}$, $p_B v_B = 3000 \times 1 = \mathbf{3000}$.

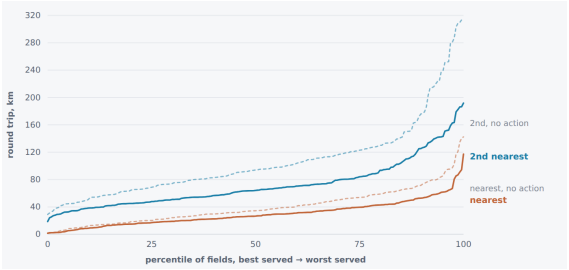
Four Objectives Lead to Four Different Answers



Unweighted total distance is $x + (100 - x) = 100$ *for every* x: the objective cannot tell the plans apart. Add population and it slams to one end; add vulnerability and it slams to the *other*; ask for the worst case and it lands in the middle.

Applied to Senegal Case

Minimize total distance



Protect the worst tail



Same 225 fields, same 32 cities, same five shops to give away. The mean barely moves; **the tail collapses.**

Who Represents What?

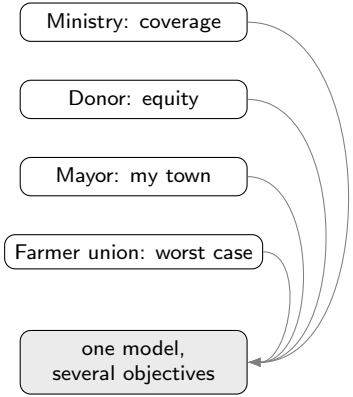
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- The model becomes a way to **structure the argument, not to end it**!
- In the first 15 years of work in this area, it has always been a pain to be the mathematical designer + programmer + graphic designer + ...



Visualization

(Visualization with Python and HTML by Claude)